

Numerical Methods for Differential Equations

Projects Report

July 21, 2026

Contents

1	Project 1: Poisson equation	2
1.1	Problem Formulation	2
1.2	Solution of the problem	2
1.3	Implementation of Galerkin $\mathcal{P}^1$	3
1.3.1	Stiffness Matrix and Right-Hand Side Assembly	3
1.4	Imposing Boundary Conditions	4
1.4.1	Error Evaluation Formulation	4
1.5	Numerical Results and Visualizations	5
1.5.1	Plot of U and Uh	5
1.5.2	Plots of error surface	6
1.6	Experimental Convergence and Error Analysis	6
2	Project 2: Convection Diffusion equation	8
2.1	Problem Formulation and Galerkin Approach	8
2.2	Implementation without stabilization	8
2.2.1	pseudo Peclet number	9
2.3	Implementation of SUD	10
2.4	Stabilization results	10
3	project 3: Parabolic equation	11
3.1	Problem Formulation	11
3.2	Implementation of solver	12
3.2.1	Error Estimates and Expected Convergence	13
3.3	Test of Δt : Numerical Results	14
3.3.1	$\theta = 1, 1/2$	14
3.3.2	$\theta = 0$	15

1 Project 1: Poisson equation

1.1 Problem Formulation

In the first project we consider the Poisson equation:

$$-\Delta u = f \quad \text{in } \Omega \subset \mathbb{R}^2 \quad (1)$$

Which is a second order divergence form elliptic equation with diffusion coefficient $D = 1$ and parameter $\lambda = 0$. We consider the domain

$$\Omega = [-1, 1]^2 = \{(x, y) \in \mathbb{R}^2 : -1 \leq x, y \leq 1\},$$

and Dirichlet and Neumann boundary conditions

$$u = 0 \quad \text{on } \Gamma_D = \{(x, y) \in \mathbb{R}^2 : -1 \leq x \leq 1, y = \pm 1\} \quad (2)$$

$$\nabla u \cdot \nu = q_1 = -\pi \cos\left(\frac{\pi}{2}y\right) \quad \text{on } \Gamma_{N_1} = \{(x, y) \in \mathbb{R}^2 : x = 1, -1 \leq y \leq 1\} \quad (3)$$

$$\nabla u \cdot \nu = q_2 = \pi \cos\left(\frac{\pi}{2}y\right) \quad \text{on } \Gamma_{N_2} = \{(x, y) \in \mathbb{R}^2 : x = -1, -1 \leq y \leq 1\} \quad (4)$$

For this problem, we are provided the explicit solution as

$$u(x, y) = \sin(\pi x) \cos\left(\frac{\pi}{2}y\right)$$

Note:

- The solution in $C^\infty(\Omega)$ (and in particular $H^2(\Omega)$)
- Ω is bounded lipschitz and convex
- We have mixed BCs

1.2 Solution of the problem

Within this project, given a fixed mesh (triangulation) $\mathcal{T}_h(\Omega)$ we shall do a numerical approximation using the Galerkin $\mathcal{P}^1$ method by the following scheme:

Find $u_h \in \mathcal{V}_h = \text{span}\{\phi_1, \dots, \phi_n\}$ such that

$$a(u_h, \phi_j) = (f, \phi_j) + (q_n, \phi_j)_{\Gamma_N} \quad \forall j = 1, \dots, n \quad (5)$$

where

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx; \quad (f, v) = \int_{\Omega} f v \, dx; \quad (q_n, \phi_i)_{\Gamma_N} = \int_{\Gamma_N} q_n \phi_i \, ds.$$

To find u_h , we expand it in terms of the basis functions of $\mathcal{V}_h$

$$u_h(x, y) = \sum_{i=1}^n u_i \phi_i(x, y)$$

where $\phi_i(x, y)$ are the piecewise linear Lagrangian basis functions, and the coefficients u_i are the unknown values of the approximate solution at the n nodal points of the triangulation, such that $u_i = u_h(x_i)$. Because the chosen basis functions are continuous across the elements, u_h is globally continuous over the domain by construction.

By bilinearity of a , we can find u_h by transforming (5) into a linear system of equation

$$Au = b \quad (6)$$

where $A = (a_{ij})$ is the stiffness matrix defined by $a_{ij} = a(\phi_i, \phi_j)$, b as the load vector $b_i = (f, \phi_i) + (g, \phi_i)_{\Gamma_{N_h}}$, and u is a column vector consisting nodal points of u_h .

Within Galerkin $\mathcal{P}^1$ scheme, the basis of $\mathcal{V}_h$ is given by conforming piecewise linear two-dimensional interpolating polynomials defined element-by-element. Hence, the (i, j) -th component a_{ij} is provided by assembly of local contributions of $a_{ij}^{(T)}$ from $\phi_i^{(T)}$ and $\phi_j^{(T)}$ which are Langarians that have node (x_i, y_i) , (x_j, y_j) as vertexes of triangle T :

$$a_{ij} = \sum_{T \in \mathcal{T}_h(\Omega)} a_{ij}^{(T)} = \sum_{T \in \mathcal{T}_h(\Omega)} \int_T \nabla \phi_i^{(T)} \cdot \nabla \phi_j^{(T)} dx dy$$

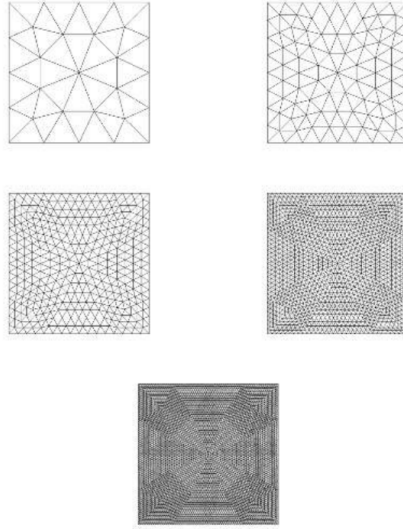
while the right hand side b is given by:

$$b_i = \sum_{T \in \mathcal{T}_h(\Omega)} \int_T f(x, y) \phi_i^{(T)} dx dy + \sum_{e \in \Gamma_{N_h}} \int_e q(s) \phi_i^{(T)} ds$$

where e indexes the triangle edges that belong to the Neumann boundary $\Gamma_{N, h}$.

1.3 Implementation of Galerkin $\mathcal{P}^1$

The implementation is evaluated across a shape regular sequence of five uniformly refined meshes, with the characteristic length parameter scaling as $h = 0.5 \times 2^{-i}$ for $i = 0, \dots, 4$.



1.3.1 Stiffness Matrix and Right-Hand Side Assembly

The global matrix A is assembled element-by-element. For each triangle $T \in \mathcal{T}_h(\Omega)$, the local affine shape functions take the form $\phi_i^{(T)} = a_i^{(T)} + b_i^{(T)}x + c_i^{(T)}y$. The exact entries of the local stiffness matrix are computed analytically as:

$$a_{ij}^{(T)} = \int_T \nabla \phi_i^{(T)} \cdot \nabla \phi_j^{(T)} dx dy = \text{area}(T) \cdot (b_i^{(T)} b_j^{(T)} + c_i^{(T)} c_j^{(T)}). \quad (7)$$

Simultaneously, the right-hand side load vector b is approximated using the vertex-based trapezoidal quadrature rule for the internal domain contributions:

$$b_i \approx \sum_{T \in \mathcal{T}_h(\Omega)} \frac{\text{area}(T)}{3} f(x_i, y_i). \quad (8)$$

1.4 Imposing Boundary Conditions

Neumann Conditions: Within this problem, we have two Neumann boundaries. Even though the Neumann flux on Γ_{N_1} and Γ_{N_2} are written differently, we can unify them as:

$$q(x, y) = -\text{sign}(x)\pi \cos\left(\frac{\pi}{2}y\right)$$

We shall update the load vector for Neumann nodes by evaluating the line integral over the boundary edges $e \in \Gamma_{N,h}$. Using the 1D trapezoidal rule, we add:

$$b_{i,Neu} = \sum_{e \in \Gamma_{N,h}} \int_e q(s)\phi_i(s) ds \approx \sum_{e \in \Gamma_{N,h}} \frac{\ell_e}{2} \left(q(x_1^{(e)}, y_1^{(e)})\phi_i(x_1^{(e)}, y_1^{(e)}) + q(x_2^{(e)}, y_2^{(e)})\phi_i(x_2^{(e)}, y_2^{(e)}) \right)$$

where $(x_1^{(e)}, y_1^{(e)})$ and $(x_2^{(e)}, y_2^{(e)})$ are the endpoints of the boundary edge e . Some work has been done in order to compute the exact length of all the edges, considering also those who shared a Dirichlet node.

Dirichlet Conditions (Penalty Method): To strictly enforce the homogeneous Dirichlet condition $u_h(x_i, y_i) = 0$ on Γ_D , we utilize the penalty method. For any Dirichlet node i , the diagonal entry of the global stiffness matrix is overwritten with a large penalty parameter $\lambda = \frac{1}{\epsilon}$ (choosing $\epsilon = 10^{-15}$). The corresponding entry in the load vector is set to $\frac{1}{\epsilon}g(x_i, y_i) = 0$ (we are saying $1e15 * u1 + A1, 2 * u2 + \dots + A1, nodes * u_{Nodes} = 1e15 * b1$ which approximatively implies $u1=b1$). As $\epsilon \rightarrow 0$, this effectively decouples the i -the equation to exactly force $u_i = 0$ without altering the matrix dimensions.

1.4.1 Error Evaluation Formulation

To verify theoretical convergence, the global errors $e_h := u - u_h$ are evaluated element-by-element. The L^2 -norm is approximated via the vertex trapezoidal rule:

$$\|e_h\|_{L^2(T)}^2 = \int_T |u - u_h|^2 dx dy \approx \frac{\text{area}(T)}{3} \sum_{i=1}^3 \left| u(x_i^{(T)}, y_i^{(T)}) - u_h(x_i^{(T)}, y_i^{(T)}) \right|^2. \quad (9)$$

Because the numerical gradient ∇u_h is constant inside the $\mathcal{P}^1$ element, the H^1 -seminorm is optimally evaluated using the one-point midpoint quadrature rule at the barycenter $(x_b^{(T)}, y_b^{(T)})$:

$$|e_h|_{H^1(T)}^2 = \int_T |\nabla u - \nabla u_h|^2 dx dy \approx \text{area}(T) \left| \nabla u(x_b^{(T)}, y_b^{(T)}) - \nabla u_h \right|^2. \quad (10)$$

The global error norms are then computed by summing the squared local contributions before taking the square root:

$$\begin{aligned} \|e_h\|_{L^2(\Omega)} &= \sqrt{\sum_{T \in \mathcal{T}_h(\Omega)} \|e_h\|_{L^2(T)}^2} \\ \|e_h\|_{H^1(\Omega)} &= \sqrt{\|e_h\|_{L^2(\Omega)}^2 + \sum_{T \in \mathcal{T}_h(\Omega)} |e_h|_{H^1(T)}^2} \end{aligned}$$

Following the computation of the global errors across uniformly refined meshes, the experimental order of convergence (EOC) between a mesh parameter h and its refinement $h/2$ can be estimated by:

$$p \approx \log_2 \left(\frac{\|e_h\|}{\|e_{h/2}\|} \right)$$

1.5 Numerical Results and Visualizations

1.5.1 Plot of U and U_h

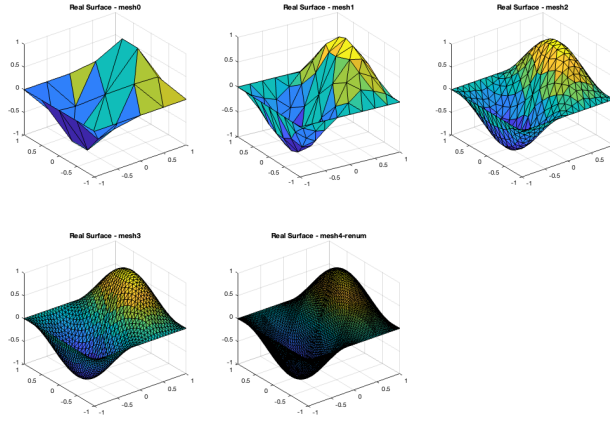


Figure 1: graphics of the real surface along the meshes

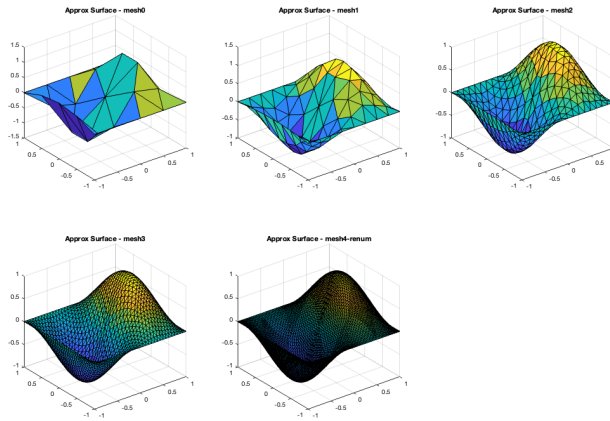


Figure 2: graphics of the approximated surface along the meshes

The numerical approximation $u_h(x, y)$ becomes smoother as the meshes refine and follows the exact analytical solution across the domain. We can notice that fluctuations exist along the Neumann boundaries, although they tend to vanish upon mesh refinement. Furthermore, the penalty method successfully and strictly maintains $u_h = 0$ at the Dirichlet boundaries ($y = \pm 1$) across all mesh levels.

1.5.2 Plots of error surface

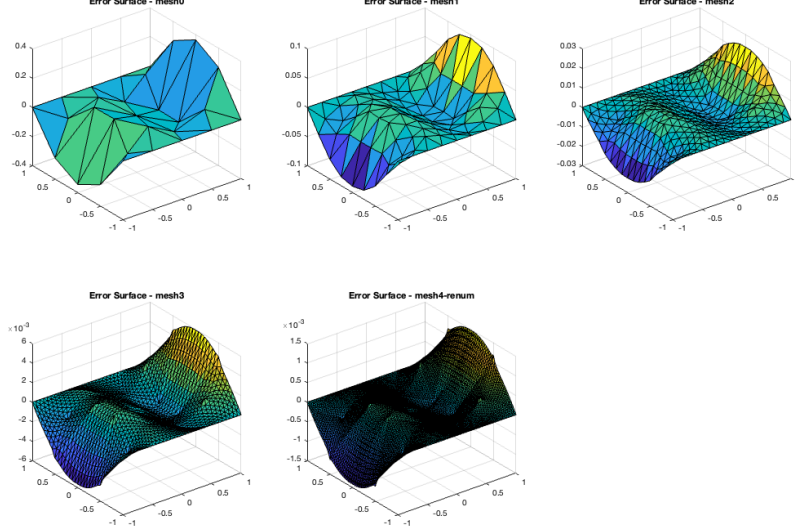


Figure 3: graphics of the approximated surface along the meshes

Analysis of the pointwise error distribution $e_h(x, y)$ reveals that the largest deviations are naturally concentrated around the Neumann boundaries and the steep gradient regions of the exact solution, scaling down proportionally as expected with h .

1.6 Experimental Convergence and Error Analysis

The assumptions on our problem allows the convergence estimates seen in class to work. According to Céa's lemma, interpolation estimates, and the Aubin-Nitsche duality argument for the L^2 norm, the theoretical a priori error estimates for $\mathcal{P}^1$ finite elements are:

$$\|u - u_h\|_{H^1(\Omega)} \leq Ch|u|_{H^2(\Omega)} = \mathcal{O}(h), \quad (11)$$

$$\|u - u_h\|_{L^2(\Omega)} \leq Ch^2|u|_{H^2(\Omega)} = \mathcal{O}(h^2). \quad (12)$$

Consequently, we anticipate an asymptotic error reduction ratio (e_{2h}/e_h) of exactly 4 for the L^2 -norm, and 2 for the H^1 -seminorm. The experimental measurements across the successive uniform mesh refinements are tabulated in Table 1.

Table 1: Experimental convergence for the Poisson equation. The ratios strictly approach the theoretical expectations of 4.0 for L^2 and 2.0 for H^1 .

Mesh Level	Parameter (h)	L^2 Error Analysis		H^1 Semi-norm Analysis	
		Norm ($\ e_h\ _{L^2}$)	Ratio (e_{2h}/e_h)	Norm ($ e_h _{H^1}$)	Ratio (e_{2h}/e_h)
Mesh 0	7.11×10^{-1}	2.5124×10^{-1}	—	1.3309×10^0	—
Mesh 1	3.56×10^{-1}	5.1095×10^{-2}	4.9171	5.2703×10^{-1}	2.5253
Mesh 2	1.78×10^{-1}	1.1994×10^{-2}	4.2601	2.4361×10^{-1}	2.1634
Mesh 3	8.90×10^{-2}	2.9470×10^{-3}	4.0699	1.1919×10^{-1}	2.0438
Mesh 4	4.44×10^{-2}	7.3346×10^{-4}	4.0179	5.9269×10^{-2}	2.0111

The computed slopes of the least-squares best fit lines in a log-log scale plot are approximately 2.10 for the L^2 -norm and 1.11 for the H^1 -seminorm. Examining the terminal ratios (4.0179 and 2.0111) definitively confirms that the numerical scheme has reached its asymptotic convergence regime, perfectly validating the theoretical a priori estimates derived for the Galerkin formulation.

